TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from ……../20…. To ……../20…..

(*) Profession: <Software Engineer> **Specialty**: <ES> [X] <IS> [ ] <JS> [ ]

(*) Kinds of person make registers: Lecturer [X] Students [ ]

1. Register information for supervisor (if have)

No.	Fullname	Phone	**E-Mail**	**Title**
Supervisor	Đỗ Tấn Nhàn	0903056041	nhandt35@fe.edu.vn	Mr.

2. Register information for students (if have)

	Full name	**Student code**	**Phone**	**E-mail**	**Role in Group**
Student 1	Huỳnh Kim Phúc	SE192882	0902477473	kphuchuynh2005@gmail.com	fe-leader
Student 2	Nguyễn Trần Minh Hưng	SE193002	0767090509	minhhung19052018@gmail.com	fe
Student 3	Nguyễn Trần Quang Vinh	SE180354	0934812879	vinhntqse180354@fpt.edu.vn	be
Student 4	Huỳnh Tuấn Kiệt	SE180309	0949564231	kiethtse180309@fpt.edu.vn	be
Student 5	Đinh Hoàng Kha	SE193633	0964901700	dinhhoangkha107@gmail.com	fe

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

English: **Intelligent Hotel Concierge Robot**
Vietnamese: Robot trợ lý dịch vụ khách sạn thông minh
Abbreviation: HCRobot

a. Context:

The hospitality industry increasingly emphasizes delivering personalized and efficient guest services to enhance customer experience. However, during peak periods, hotel

staff are often required to handle multiple inquiries and service requests simultaneously, leading to increased workload and longer response times. Although chatbot-based solutions can provide basic information, they lack physical presence and seamless collaboration with hotel staff. Therefore, there is a growing need for an intelligent concierge robot that can assist guests, provide service recommendations, and support hotel staff in delivering consistent and high-quality guest services.

b. Proposed Solutions

Intelligent Hotel Concierge Robot is proposed as a virtual concierge and guest service assistant designed to improve guest experience and support hotel operations. The robot provides multilingual interaction, delivers information about hotel facilities, amenities, and nearby attractions, and recommends services based on guests' needs. It also receives and manages guest service requests, such as housekeeping and transportation assistance, and collaborates with hotel staff through a human-in-the-loop mechanism when human intervention is required. By integrating conversational AI and guest service support, the robot aims to enhance service efficiency and reduce staff workload.

c. Functional requirement

- **Module for Guest**
  - Interact with the robot through voice commands or the display screen and select the preferred language.
  - Search for information about hotel facilities, amenities, promotions, and nearby attractions.
  - Receive recommendations for hotel services, dining options, transportation, and local points of interest.
  - Submit guest service requests such as housekeeping, additional amenities, wake-up calls, or transportation assistance.
  - Track the status of submitted requests.
  - Request assistance from hotel staff when necessary.
  - Rate the quality of the robot's assistance after completing an interaction.

- **Module for Robot**
  - Detect guests, greet them, and initiate interactions automatically.
  - Communicate with guests in multiple languages through voice and display interfaces.
  - Act as a virtual concierge by providing information about hotel facilities, amenities, policies, and nearby attractions.
  - Recommend hotel services, dining options, transportation, and local points of interest based on guests' needs.
  - Maintain conversation context and provide personalized interactions.

- Receive, categorize, and transfer guest service requests to hotel staff when necessary.
- Assist communication between international guests and hotel staff.
- Display announcements, promotions, and hotel information while in idle mode.
- Perform simple expressions and responses to enhance human-robot interaction.
- Record conversation history and guest service activities.
- Monitor its operating status and generate alerts when abnormal conditions occur.

- **Module for Hotel Staff**
  - Receive and handle guest requests transferred by the robot.
  - Manage and update the status of guest service requests.
  - Provide personalized assistance when requests require human intervention.
  - Monitor ongoing guest interactions and the robot's operating status.
  - Receive alerts for requests or situations requiring immediate attention.
  - Review conversation history and guest service records.
  - Update response content and knowledge to improve the robot's service quality.

- **Module for Admin**
  - Manage user accounts and access permissions.
  - Manage hotel information, amenities, promotions, and content displayed by the robot. (hotel content tab)
  - Manage the knowledge base and guest service content. (knowledge tab)
  - Manage robot profiles, operating status, activity history, and maintenance schedules.
  - Monitor statistics related to guest interactions, service requests, and guest satisfaction.
  - Manage conversation logs, activity records, and system reports.
  - Configure system settings and operational parameters.

d. Non-functional requirement:
   - The system is capable of keeping track of the user at the time when the user performs important processing functions in the system.
   - Unicode font.

(*) 3.2. Main proposal content (including result and product)

a. Theory and practice (document):

- The documents include User Requirement, Software Requirement Specification, Architecture Design, Detail Design, Testing Document, Installation Guide, User Guide, Sources Code.
- Server-side technologies:
  - Server: .NET, Windows Azure/AWS, …
  - Database Design: SQL Server, MySQL, …
- Client-side technologies:
  - Web Client: HTML5, CSS3, Typescript, ReactJS, ...
  - Mobile App: Flutter, Xamarin, …

b. Products:

- Mobile/Web App for Guest.
- Robot Application.
- Mobile/Web App for Hotel Staff.
- Web App for Admin.

c. Proposed Tasks:

- Task package 1: Develop the application.
- Task package 2: Build and Test the system.
- Task package 3: Prepare all the required documents.

4. Other comments (propose all relative things if have).

	HCM, date …… ……. /20 …
Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*

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